

$[2r - 1]_t$ -divisibility of S_r -isotypic multiplicities of $R_{(k^r)}$ for arbitrary k : the Molien bound suffices

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Abstract

Fix integers $r \geq 2$, $k \geq 2$ and set $n = rk$, $d = 2r - 1$, $\alpha = (k^r)$. Let $R_\alpha = R_n^{S_\alpha}$ be the parabolic coinvariant algebra of S_n (a graded S_r -module via the residual block-permutation action of $H/S_\alpha \cong S_r$), and let $m_\pi(t) = \sum_\lambda \tilde{f}_\lambda(t) \langle s_\pi[h_k], s_\lambda \rangle$ be the graded S_r -isotypic multiplicities (Theorem D). We prove:

Theorem. *Assume $d = 2r - 1$ is prime and $k \bmod d \in \{2, 3, \dots, d - 1\}$. Then*

$$[d]_t \mid m_\pi(t) \quad \text{for every } \pi \vdash r.$$

This subsumes and strengthens the “first-period” theorem of [1] (which restricted to $2 \leq k \leq d - 1$). The proof is a single-line refinement of the Molien-Springer argument of [1]: the previously-used inequality $m_d(\tilde{\sigma}h) = 0$ (uniform term-by-term vanishing) is replaced by the weaker $m_d(\tilde{\sigma}h) < \lfloor n/d \rfloor$, which suffices for the trace $\text{tr}(\tilde{\sigma}h \mid R_n; t)$ to vanish at ζ_d . The weaker inequality holds throughout the full “allowed” range $k \bmod d \in \{2, \dots, d - 1\}$, so *term-by-term* vanishing extends to arbitrarily large k ; no cancellation across h is needed.

The theorem covers every empirically verified positive case with prime $d = 2r - 1$ (through $r = 6$; see [~/projects/probes/2026-07-31-2r-1-divisibility/](#)). It is verified computationally for the smallest new cases $(r, k) \in \{(3, 7), (3, 8), (3, 9), (3, 12), (4, 9)\}$ in §5.

1 Setup and reduction to characters

We recall the notation and reductions of [1].

Let S_n act on the polynomial ring $\mathbb{C}[x_1, \dots, x_n]$ by permutation and let $R_n = \mathbb{C}[x_1, \dots, x_n]/(\mathbb{C}[x_1, \dots, x_n]_+^{S_n})$ be the coinvariant algebra. For $\alpha = (k^r)$, $n = rk$, let $S_\alpha = S_k^r \subseteq S_n$, $H = S_k \wr S_r$ its wreath normaliser, and $R_\alpha = R_n^{S_\alpha}$ the parabolic coinvariant algebra. The residual quotient $H/S_\alpha \cong S_r$ acts on R_α by graded ring automorphisms ([2, Thm 1(i)]).

For each $\pi \vdash r$, the graded π -isotypic multiplicity is $m_\pi(t) = \sum_i t^i \dim \text{Hom}_{S_r}(V_\pi, (R_\alpha)_i)$; Theorem D of [2] gives the closed form $m_\pi(t) = \sum_\lambda \tilde{f}_\lambda(t) \langle s_\pi[h_k], s_\lambda \rangle$. For $\sigma \in S_r$, write

$$P_\sigma(t) := \sum_{\pi \vdash r} \chi^\pi(\sigma) m_\pi(t)$$

for the graded character of σ on R_α ; by orthogonality,

$$m_\pi(t) = \frac{1}{r!} \sum_{\sigma \in S_r} \chi^\pi(\sigma) P_\sigma(t).$$

Since d is prime, $[d]_t = \Phi_d(t)$ is irreducible over $\mathbb{Z}[t]$, so $[d]_t \mid m_\pi(t)$ iff $m_\pi(\zeta_d) = 0$, where $\zeta_d = e^{2\pi i/d}$.

Lemma 1 ([1, Lem 4.1]). $m_\pi(\zeta_d) = 0$ for every $\pi \vdash r$ if and only if $P_\sigma(\zeta_d) = 0$ for every $\sigma \in S_r$.

The theorem thus reduces to the following.

Proposition 2. For $r \geq 2$, $d = 2r - 1$ prime, and $k \bmod d \in \{2, \dots, d - 1\}$: $P_\sigma(\zeta_d) = 0$ for every $\sigma \in S_r$.

2 The Molien bound

For $\sigma \in S_r$ we write $\tilde{\sigma} \in S_n$ for its block-permutation lift, viewing $S_r \hookrightarrow N_{S_n}(S_\alpha)/S_\alpha$.

Lemma 3 (Molien formula for P_σ ; [1, Lem 4.3]). For every $\sigma \in S_r$,

$$P_\sigma(t) = \frac{1}{|S_\alpha|} \sum_{h \in S_\alpha} \frac{\prod_{i=1}^n (1 - t^i)}{\prod_j (1 - t^{c_j(\tilde{\sigma}h)})},$$

where the inner product ranges over cycles of $\tilde{\sigma}h$ as an element of S_n .

Lemma 4 (Vanishing order of individual terms; [1, Lem 4.4]). For any $g \in S_n$, let $m_d(g) = \#\{j : d \mid c_j(g)\}$. Then $\text{tr}(g \mid R_n; t) = \prod_{i=1}^n (1 - t^i) / \prod_j (1 - t^{c_j(g)})$ has a zero of order exactly $\lfloor n/d \rfloor - m_d(g)$ at $t = \zeta_d$. In particular, $\text{tr}(g \mid R_n; \zeta_d) = 0$ iff $m_d(g) < \lfloor n/d \rfloor$.

Lemma 5 (Wreath cycle formula; [1, Lem 4.5], [3, Ch. I App. B]). Let $\sigma \in S_r$, $h = (h_1, \dots, h_r) \in S_\alpha = S_k^r$, and $\tilde{\sigma}h \in S_n$. For each cycle $(b_1 \ b_2 \ \dots \ b_\ell)$ of σ , form the companion permutation $g_c := h_{b_\ell} h_{b_{\ell-1}} \dots h_{b_1} \in S_k$. Then the cycles of $\tilde{\sigma}h$ on the ℓk elements of blocks $b_1, \dots, b_\ell$ have lengths precisely $\{\ell m : m \text{ is a cycle length of } g_c\}$.

We now prove the main bound.

Proposition 6 (Main bound). Let $r \geq 2$, $d = 2r - 1$ prime, $k \geq 2$, and $\sigma \in S_r$ with $p = p(\sigma)$ cycles (counting fixed points). Then

$$\max_{h \in S_\alpha} m_d(\tilde{\sigma}h) \leq p \cdot \lfloor k/d \rfloor.$$

Proof. By Lemma 5, every cycle of $\tilde{\sigma}h$ has length ℓm for some cycle length ℓ of σ (with $\ell \leq r$) and some cycle length m of the corresponding companion $g_c \in S_k$. Since $d = 2r - 1 > r \geq \ell$ and d is prime, $\gcd(\ell, d) = 1$; hence $d \mid \ell m$ if and only if $d \mid m$. Consequently

$$m_d(\tilde{\sigma}h) = \sum_{\text{cycles } C \text{ of } \sigma} m_d(g_c(C)),$$

where $g_c(C) \in S_k$ is the companion of cycle C .

For any $g \in S_k$, cycles of g of length divisible by d each use at least d elements out of k , so $m_d(g) \leq \lfloor k/d \rfloor$. Summing over the p cycles of σ gives the claim. $\square$

Corollary 7. Write $k = ad + b$ with $a = \lfloor k/d \rfloor$ and $b = k \bmod d$. Then for every $\sigma \in S_r$ with p cycles and every $h \in S_\alpha$,

$$\lfloor n/d \rfloor - m_d(\tilde{\sigma}h) \geq (r - p)a + \lfloor rb/d \rfloor.$$

Proof. $n = rk = rad + rb$ gives $\lfloor n/d \rfloor = ra + \lfloor rb/d \rfloor$ (the division is exact on rad ; the leftover rb contributes $\lfloor rb/d \rfloor$). Subtract the bound of Proposition 6. $\square$

3 Proof of Proposition 2 and the theorem

Proof of Proposition 2. Assume $b := k \bmod d \in \{2, \dots, d-1\}$. We show that $\lfloor n/d \rfloor - m_d(\tilde{\sigma}h) \geq 1$ for every $\sigma \in S_r$ and $h \in S_\alpha$, whence every summand of the Molien formula of Lemma 3 vanishes at ζ_d by Lemma 4.

By Corollary 7, the gap is at least $(r-p)a + \lfloor rb/d \rfloor$ where $p = p(\sigma)$ and $a = \lfloor k/d \rfloor$.

Case 1: $\sigma = e$. Then $p = r$, and the gap is at least $\lfloor rb/d \rfloor$. Since $b \geq 2$, we have $rb \geq 2r = d+1 > d$, hence $\lfloor rb/d \rfloor \geq 1$.

Case 2: $\sigma \neq e$. Then σ has at least one cycle of length ≥ 2 , so $p \leq r-1$; hence $r-p \geq 1$. Since $a \geq 0$ and $\lfloor rb/d \rfloor \geq 1$ (as in Case 1),

$$(r-p)a + \lfloor rb/d \rfloor \geq 0 + 1 = 1.$$

In either case, the gap is ≥ 1 , so $P_\sigma(\zeta_d) = 0$. □

Proof of the theorem. Combine Proposition 2 with Lemma 1 and the equivalence $[d]_t \mid f(t) \iff f(\zeta_d) = 0$ (valid since d prime, $[d]_t = \Phi_d(t)$). □

4 Discussion

Remark (Order of vanishing). Corollary 7 in fact shows that every summand of Lemma 3 vanishes at ζ_d to order at least $(r-p)a + \lfloor rb/d \rfloor$. For $\sigma = e$ this is $\lfloor rb/d \rfloor$, so $P_e(t) = \binom{n}{k^r}_t$ vanishes at ζ_d to order at least $\lfloor rb/d \rfloor$. This is in fact the exact order of vanishing of $\binom{n}{k^r}_t$ at ζ_d (a direct calculation from $\binom{n}{k^r}_t = [n]_t! / ([k]_t!)^r$: the order is $\lfloor n/d \rfloor - r\lfloor k/d \rfloor = \lfloor rb/d \rfloor$). So Corollary 7 is sharp at $\sigma = e$; term-by-term vanishing is not enough to detect any *cancellation*, which for $\sigma = e$ is empty by definition. For $\sigma \neq e$ the bound may be conservative.

Remark (Why the first-period theorem restricted to $k < d$). The first-period argument of [1] used the stronger inequality $m_d(\tilde{\sigma}h) = 0$ (its Cor. 4.5), obtained from $k \leq d-1$: then no cycle of any companion has length divisible by d , so trivially $m_d = 0$. For $k \geq d$ this is false: taking $\sigma = e$ and $h_i = (12 \cdots d)(d+1) \cdots (k)$ gives $m_d(h) = r \geq 1$. But we do not need $m_d = 0$; the weaker $m_d < \lfloor n/d \rfloor$ suffices, and Proposition 6 shows the weaker inequality throughout the allowed range.

Remark (Complementary sharpness at forbidden residues). When $b \in \{0, 1\}$ (the “forbidden” residues), the bound of Proposition 6 is achieved with equality at $\sigma = e$: taking each $h_i \in S_k$ with cycle type $d^a \cdot 1^b$ gives $m_d(h) = ra$; and $\lfloor n/d \rfloor - ra = \lfloor rb/d \rfloor$. For $b = 0$: gap = 0, so the $\sigma = e$ Molien term does *not* vanish and there is no divisibility (matching the empirical pattern). For $b = 1$: $\lfloor r/d \rfloor = 0$ (since $r < d$), so gap = 0 again. Thus the same argument gives a clean explanation for the “forbidden” residues without invoking cancellation.

Remark (Composite d). The prime hypothesis on d is used at two points: (i) to reduce $[d]_t \mid f$ to $f(\zeta_d) = 0$; (ii) to force $d \mid \ell m \Rightarrow d \mid m$ for $\ell \leq r < d$ prime. For composite $d = 2r-1$ (e.g. $r = 5, d = 9$), point (i) becomes divisibility by each cyclotomic factor $\Phi_e(t)$, $e \mid d$, separately; point (ii) requires $e > r$ for each prime factor e , which fails when r has a small prime factor $< d$. Thus the argument as stated does not apply.

Remark (Isotypic vs. Hilbert-series interpretation). Setting $\sigma = e$ gives the well-known cyclotomic divisibility of the q -multinomial: $[d]_t \mid \binom{rk}{k, k, \dots, k}_t$ iff $b \geq 2$ (equivalently, $b \neq 0, 1$). This is

a classical fact and can be proved directly from $\binom{rk}{k^r}_t = [rk]_t! / ([k]_t!)^r$ by counting cyclotomic factors of $[m]_t!$ at ζ_d . The content of the present theorem is the *isotypic* refinement: the same factor $[d]_t$ divides every $m_\pi(t)$, not merely their f^π -weighted sum $\binom{rk}{k^r}_t = \sum_\pi f^\pi m_\pi(t)$.

5 Verification

The Molien bound of Proposition 6 was verified by brute enumeration of $S_\alpha = S_k^r$ for $(r, k) \in \{(2, 5), (3, 2)\}$, computing $\max_h m_d(\tilde{\sigma}h)$ for each conjugacy class of $\sigma \in S_r$ and checking against the theoretical bound $p[k/d]$. In every case the bound matches the enumerated maximum.

The multinomial vanishing $\binom{n}{k^r}(\zeta_d) = 0$ (i.e., the $\sigma = e$ case) was verified symbolically for $(r, k) \in \{(2, 5), (3, 7), (3, 8), (3, 9), (3, 12), (4, 9)\}$.

The full isotypic divisibility $[d]_t \mid m_\pi(t)$ was verified by explicit computation of $m_\pi(t)$ (via Theorem D expansion plus fake degrees) for $(r, k, d) = (3, 7, 5)$ and each $\pi \in \{(3), (2, 1), (1, 1, 1)\}$; in every case $[5]_t \mid m_\pi(t)$. Scripts: `~/projects/probes/2026-08-04-prove-second-period/verify.py`.

References

- [1] C. Muse (Clio), *$[2r - 1]_t$ -divisibility of S_r -isotypic multiplicities of $R_{(k^r)}$ for $2 \leq k \leq 2r - 2$: a Molien-Springer proof*, note (2026-07-31).
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- [3] I.G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford Univ. Press, 1995.